

Summary of Findings for Data Analyses Across Counties (Copiah, Claiborne, Warren)

Part 1: Demographic profiles

The demographic profiles between Copiah, Claiborne, and Warren is the only variable that largely changes across each of these counties. Copiah is mostly white at 59.5 percent of the population, whereas Claiborne is 88.2 percent black. Warren is a more racially balanced county at 29.6 percent white and 46.4 percent black. The gender distribution is largely similar across all three counties—within 1 percent of each other. Education levels across the different counties are also very similar, with roughly the same distribution of individuals from all groups. The ‘other’ group tends to be a bit more skewed, especially in this regard, but that is due to a lower number of that population being present. Unemployment across these three demographics are all relatively similar, within a few percent of one another.

Part 2: Risk scores across demographic groups

Risk distributions across counties for all three groups are fairly consistent. However, there are differences in the risk scores assigned to each demographic group. Males and females differ in that males have a slightly higher risk than females in all counties except Claiborne. Black individuals also receive a higher risk score on average in all counties when compared to the white population. Claiborne has an average risk score of 5.91 for black individuals, which is .05 and .10 points higher than Copiah and Warren, respectively. The average risk score for the white population across all counties is 4.73 whereas black individuals across all counties average out to 5.86, a full point higher than the white average. This implies disparate treatment between black and white individuals across these demographics and that the data the algorithm was likely trained on used historical data. In the United States, using historical data that includes variables such as race—a protected class—or proxy variables that may act like protected class variables may give biased outcomes such as these.

Part 3: Judge decisions to AI risk scores

The judge decisions in all counties are all distributed evenly with respect to the other counties. Across gender, both males and females are either granted or denied bail evenly at risk scores of four and five. Unfortunately, black individuals are denied bail at a much lower risk score than white individuals. White individuals with a risk score of six or higher are denied bail, while black individuals with a score of four or higher are denied bail. This is a full two point difference between two populations that are essentially the same in education, unemployment, and financial measures. This implies that there is bias not only in the algorithm in assigning risk scores, but also in the judges’ decision making to grant or deny bail when using a risk score.

Part 4: Re-offense rates and fairness metrics

FPR (false positive rate) and FNR (false negative rate) can offer insights into what mistakes the model is making. FPR is when an individual is jailed but does not re-offend, and FNR is when an individual is granted bail but does re-offend. FPR and FNR are similar across the counties within races in respect to other race populations. However, the FNR and FPR are significantly different when compared in white and black populations. The average FNR for the black population across the three counties is 71.15 while the white population has a FNR of 14.75. This means that black individuals are over 50 percent more likely to be incorrectly jailed when they will not reoffend when compared to their white counterparts. The FPR, on the other hand, is flipped in the opposite direction. The FPR was much higher for the white

population when compared to the black population, and had a FPR of 68.12 on average compared to 16.43. This means this system is very ineffective at identifying white individuals who will re-offend when compared to black individuals. The model does a very poor job at accurately categorizing risk, which may be due in part to the inflated risk scores black individuals have as well as judge decisions which tend to jail at a two point difference between the populations. Between all counties, there is no statistical parity between different racial groups and no equal opportunity for granting or denial of bail in these groups.